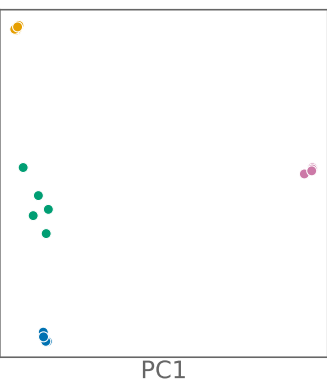
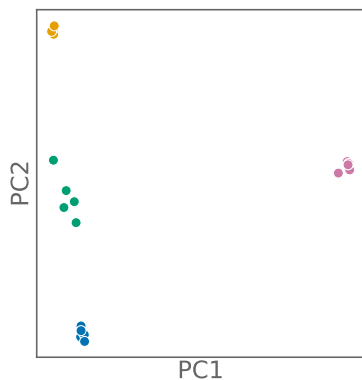


unrelated individuals only

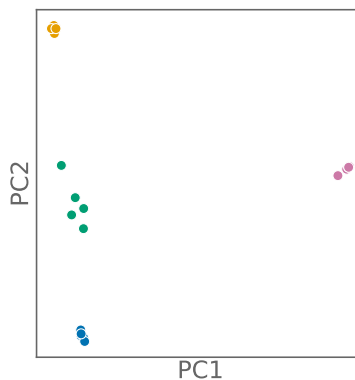
reference truth



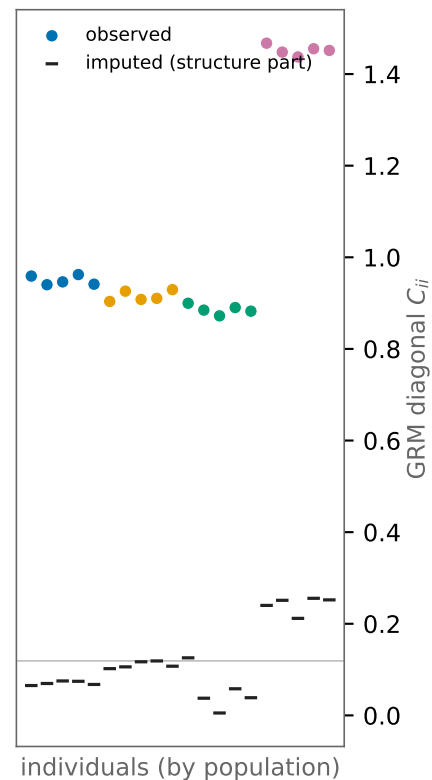
standard PCA
min $R^2 = 0.057$



standard PCA
+ --impute-diag
min $R^2 = 0.999$

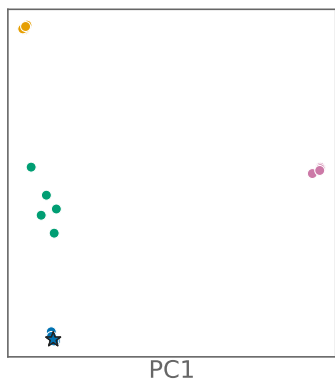


why: the observed diagonal
reflects heterozygosity

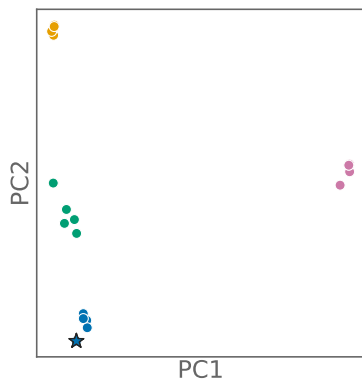


with relatives (two MZ pairs)

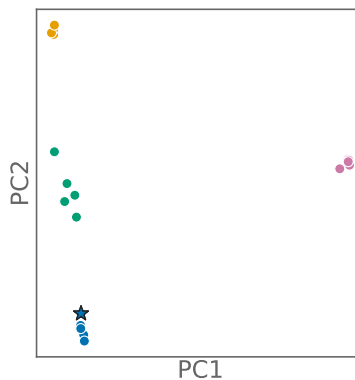
reference truth



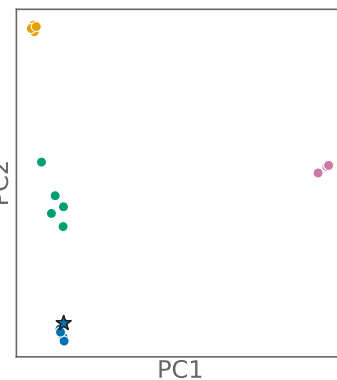
standard PCA
min $R^2 = 0.228$



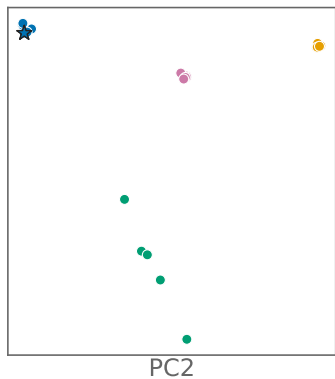
aarobust-kin
(diagonal kept)
min $R^2 = 0.038$



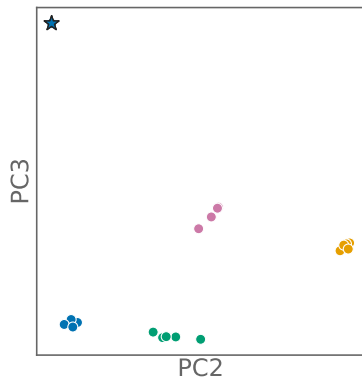
aarobust-kin
+ --impute-diag
min $R^2 = 0.999$



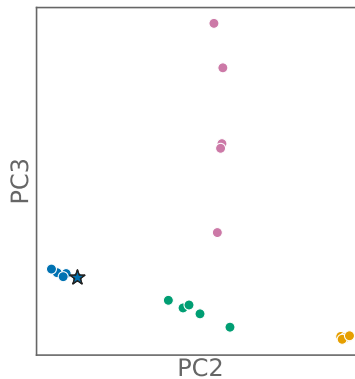
reference truth



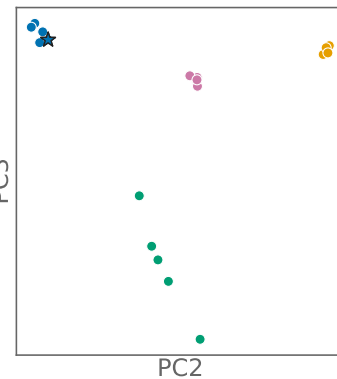
standard PCA
min $R^2 = 0.228$



aarobust-kin
(diagonal kept)
min $R^2 = 0.038$



aarobust-kin
+ --impute-diag
min $R^2 = 0.999$



● CEU ● CHB ● MXL ● YRI ★ relative